Fachhochschule Salzburg GmbH
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Software Design - Lab

Project: **Library**

Henrik Binggl Bernhard Danninger Josef Lettner

Lab Project:

In this lab you will work in teams of two to a maximum of three students and develop a small software project that will be planned, implemented, tested, and presented over the course of the semester. Each team will implement two use cases (three use cases in the case of a three-person team). Ideally, a team should consist of one designer and one developer, in order to combine different perspectives within the project work. In the project description, requirements marked with **Team (3)** are intended specifically for teams of three students.

The lab follows the typical development process of a software system. Starting with the analysis of requirements, you will create models such as use case, class, component, activity, and sequence diagrams. These models will form the blueprint for your implementation. Each team will be provided with a starter project in the chosen technology stack, which you will gradually extend by adding the designed components, persistence layer, and services.

Validation of your implementation through systematic testing is an essential part of the lab. You will create unit tests and apply mocking techniques where appropriate, ensuring that the logic of your system corresponds to the requirements specified in your models. In addition, you will apply relevant design patterns and reflect on how non-functional requirements can act as architecture drivers in your project.

Throughout the lab you will take on different roles: as analysts and designers you will create and refine models, and as developers and testers you will implement and validate them. While the implementation may sometimes follow with a certain time lag, each design activity is accompanied by coding tasks, helping you translate architectural decisions into working software and evaluate their impact in practice.

The overall aim is to deepen your understanding of the software development process by experiencing the interplay between modelling, implementation, testing, and architectural design.

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1 Procedure

When working on the individual models, it will certainly help if you are somewhat familiar with the application domain or take the time to read up on it. Most of you probably already have some experience with similar contexts — for example, having a bank account, borrowing books from a library, using a paid parking lot, or visiting a stadium. Pay attention to how certain processes are carried out in reality; this will help you approximate a workable solution. If needed, you may also do some research or ask around: for instance, what exactly happens when you purchase a ticket, borrow a book from the library, or buy shares on the stock market?

In the following sections you will find the project requirements as provided by the customer. Be aware that customers often use their own terminology, which may differ from the technical language we apply in software design. This is a typical situation in real-world projects, where part of your task as analysts and designers is to interpret and refine the given information.

As in many real projects, the initial requirements are not always fully complete and may leave room for interpretation. Try to close potential gaps in a way that seems most reasonable to you. If uncertainties remain, clarify them with the customer or stakeholders. In our lab setting, your "customer" may also be represented by the instructor, so do not hesitate to ask if questions arise.

2 Overview

We are a public library seeking to deploy a new, server-centric management solution to improve day-to-day operations and patron services.

2.1 Patron Features

The system shall allow patrons to:

- ☐ borrow (loan) library media,
- ☐ return borrowed media,
- ☐ place reservations (holds) on media,
- ☐ cancel existing reservations,
- ☐ query the availability of a specific media item,
- ☐ query the physical shelf/location of a specific media item.

If a patron exceeds the maximum loan period of four weeks, the system shall assess an overdue fee for each additional day.

2.2 Return Stations and Clients

For administration simplicity and patron convenience, the library operates multiple return stations where patrons can drop off items. Each client (web/app/kiosk/return station) operates independently and communicates with the central server.

A client shall:

- ☐ ask the server whether a specific media item may be accepted for return, and
- ☐ notify the server when a media item has been successfully returned.

The server shall respond with the current eligibility/status of the item (e.g., valid loan, item belongs to this library, item already returned) and, upon a successful return, update the item status, the patron account, and any associated reservations/holds.

2.3 Media Types

We offer the following media types:

Simple Books General-interest books such as novels and non-fiction titles.

Specialized Books Subject-specific texts focused on particular disciplines (e.g., medicine, engineering, law).

Music CDs Standard commercial audio recordings in CD format.

Movies General-audience films without age restrictions.

Magazines Periodical publications issued at regular intervals (e.g., weekly or monthly), identified by title, volume/issue, publication date, and (optionally) ISSN. Individual issues may have shorter loan periods than books, and the most recent issue may be non-circulating depending on library policy.

Movies for Adults Age-restricted films (e.g., violent or erotic content). Borrowing is limited to patrons who meet the legal age requirement; ID verification is required at checkout (and may be rechecked at return stations).

2.4 Customers

We distinguish the following customer types:

Adult Customers Regular patrons permitted to borrow all categories of media.

Children/Youth Customers (ages 10–18) Patrons who may borrow general materials but are not allowed to borrow *Movies for Adults*.

Students Patrons who may borrow *Specialized Books* for an extended loan period of up to 6 weeks. All other borrowing rules apply as usual.

3 Data Model

In software systems, the distinction between *master data* and *transactional data* plays a central role. These two concepts form the foundation for structuring and managing information in many business and application domains.

3.1 Master Data

Master data refers to core and relatively stable information that provides the basis for a wide variety of processes within a system. Such data changes infrequently compared to other types of information.

Typical examples of master data in an enterprise context include information about products, customers, suppliers, or employees. This information is essential for daily operations and often serves as reference data for other processes.

Master data provides the backbone for many processes such as sales, purchasing, or payroll. It is therefore critical that this data is consistent, accurate, and up to date, since errors or inconsistencies can affect multiple parts of a system. For this reason, master data management is often given particular attention in the design of information systems.

Note that the boundary between master data and transactional data is not always perfectly clear. In this project description, some ambiguity is intentional in order to make the requirements realistic and closer to practice.

The following statements about master data are provided for your project:

- ☐ The system shall manage library media. Each media item shall have a unique identifier and shall record a title, a description, and an age rating (FSK / age classification).
- ☐ The system shall support multiple media types with type-specific attributes:
  - ☐ **Books** (simple and specialized) : ISBN, author(s)
  - ☐ **Movies:** format, duration
  - ☐ **Audio:** codec, duration
  - ☐ **Magazines:** edition/issue
- ☐ The system shall categorize media by genres; each media item may belong to one or more genres.
- ☐ Each media item may exist in one or more physical copies.
- ☐ The system shall explicitly model library locations. A location shall consist of a shelf

identifier and one or more compartments. Each physical copy of a media item shall be assigned to exactly one compartment. Team(3)

- ☐ Each physical copy shall have its own barcode (unique copy identifier).
- ☐ The system shall maintain patron records including customer number, name, and date of birth (for age-rating checks).
- ☐ Borrowing fees/prices shall depend on the combination of media type (book, movie, etc.) and genre.

3.2 Transactional Data

Transactional data usually refers to information that records events, activities, or changes within a given context. Unlike master data, it is dynamic in nature and documents specific operations or interactions.

For example, transactional data could include customer orders, reservations, deliveries, or payments. This data changes frequently and captures the dynamic aspects of business processes.

The following statements about transactional data are provided for your project:

- ☐ The system shall allow patrons to borrow/reserve media items. A reservation shall record the reservation timestamp and a last pickup date; after the pickup deadline, the reservation shall expire automatically.
- ☐ Loans shall be created for *copies* (physical items). Each loan shall record a checkout timestamp, a planned return date, and an actual return timestamp. (Patrons reserve *media*; fulfillment occurs by loaning a specific copy of the reserved media.)
- ☐ A loan may, but need not, originate from a prior reservation. If a reservation precedes the loan, the loan record shall reference that reservation.
- ☐ The system shall enforce age-rating (FSK) restrictions; loans shall be refused when a patron does not meet the required age.
- ☐ Loans shall be time-limited.
- ☐ After the planned return date has passed, the system shall assess an overdue fee. The fee schedule shall depend on the media type and genre; if a media item is assigned to multiple genres, the highest applicable fee rate shall be used.
- ☐ For each copy, the system shall maintain usage statistics, including: number of completed loans and cumulative loan duration. The system shall support policy-based withdrawal/replacement of a copy after a configurable usage threshold is reached.

4 Use Cases

Each project member is to implement one use case within the project. The selection of use cases is **not arbitrary**; they must be chosen in the order of the list below. Implementing alternative use cases is only possible **after** consultation with the instructors.

4.1 Loan

The following business logic is to be implemented:

- ☐ The system shall allow a patron to select a *media item* (not a specific copy) to borrow.
- ☐ A loan shall be permitted only if:
  - ☐ at least one copy of the selected media is available (i.e., neither on loan nor reserved) for the requested loan period, and
  - ☐ the patron meets the age-rating (FSK) requirement.
- ☐ Upon confirmation, the system shall allocate a specific copy to the loan and record the loan metadata (checkout timestamp, planned return date).
- ☐ The system shall enforce concurrency control for multi-user operation to prevent double-allocation of the same copy.

4.2 Return

The following business logic is to be implemented:

- ☐ The return process shall be initiated by scanning the barcode of the copy.
- ☐ The system shall determine whether the planned return date has been exceeded and, if so, calculate and apply the corresponding overdue fees.
- ☐ The system shall mark the loan as closed and record the actual return timestamp.
- ☐ The system shall update usage statistics for the copy (e.g., number of loans, cumulative loan duration).
- ☐ The system shall set the copy's state to available for subsequent loans (unless blocked by policy or maintenance).

4.3 Reservation

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The following business logic is to be implemented:

- ☐ The system shall allow a patron to reserve a *media item* (not a specific copy) for a future interval.

- ☐ A reservation shall be permitted only if:
  - ☐ at least one copy is available for the requested interval (i.e., not already reserved or on loan), and
  - ☐ the patron meets the age-rating (FSK) requirement.
- ☐ When a loan is fulfilled on the basis of an existing reservation, the loan shall reference that reservation.
- ☐ The system shall enforce concurrency control for multi-user operation to prevent conflicting reservations for the same media.

5 Expected Results and Quality Assurance

By the end of the semester, each team shall deliver a working system that supports the end-to-end execution of the assigned use cases under realistic, multi-user conditions and with all relevant constraints and rules enforced. A student-facing web application shall provide the core user journey for discovering availability and placing bookings (i.e., searching and reserving items and any applicable services). In domains that involve a physical device (e.g., an ATM or a beach chair), a lightweight HTML page shall be provided to simulate the device: it must drive the same server-side flows as the real device would (for example, entering a reservation or card number, requesting unlock/deny, showing current status, and triggering state transitions).

The implementation is expected to respect concurrency and data-integrity concerns (no double booking, correct handling of overlapping requests, capacity limits, pricing or tournament rules, etc.). A bespoke CRUD user interface for master-data maintenance is *not* required; seed data, fixtures, or migration scripts are sufficient to populate master data needed for demonstrations and tests. The system shall include automated unit tests for the critical paths (reservation creation and modification, concurrent requests, device unlock/deny, capacity and rule enforcement), and the project shall document how to run these tests. During the lab we will review sound unit-testing practices (isolation, Arrange–Act–Assert, mocks/stubs, and boundary/error cases), and your test suite is expected to reflect these practices alongside basic input validation and error handling.

At submission time, each team shall hand in: (1) the implementation (source code and minimal setup/run instructions), (2) concise documentation covering the architecture, data model, and external API sufficient for another team to run and test the system, and (3) a short screencast of approximately five minutes *per use case* showing the application working in your environment and briefly explaining a code fragment that you consider important.

6 Submission

All submissions must be made exclusively via Moodle. Ensure that the names of all project members are included in every submission (e.g., on the cover page and in source code comments). **Create and maintain a single consolidated project report that you update across all deliverables. Always submit your consolidated report as a PDF file.**